CHINESE ACADEMY of MEDICAL SCIENCES OXFORD INSTITUTE

NUFFIELD DEPARTMENT of MEDICINE

UNIVERSITY OF OXFORD

Immune Recognition of Pathogens via Antigen Processing & Presentation

CAMS-COI Immunology Course October 2026

Prof Benedikt Kessler

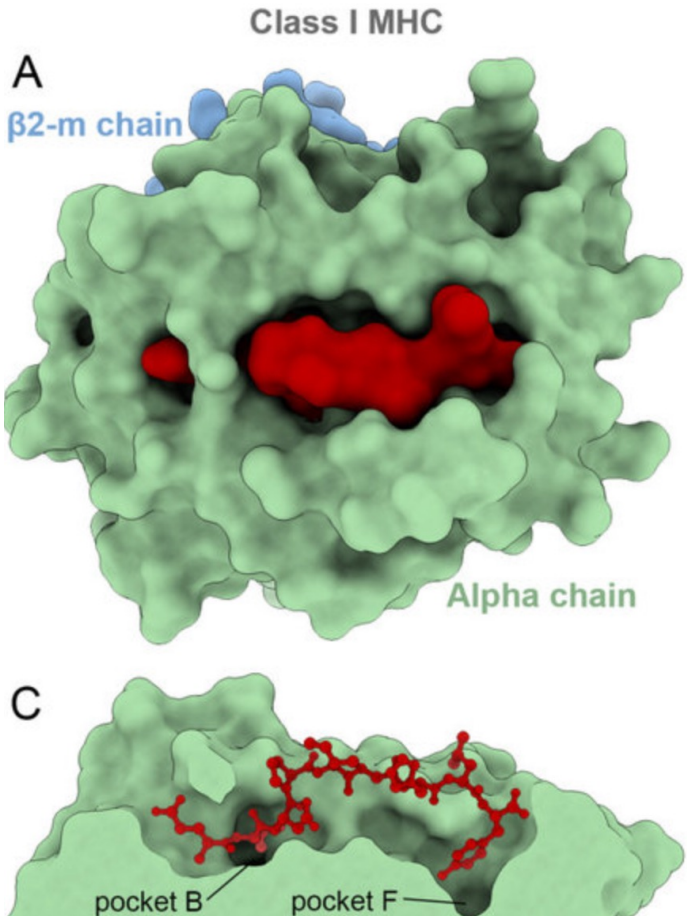
What will be covered in this lecture

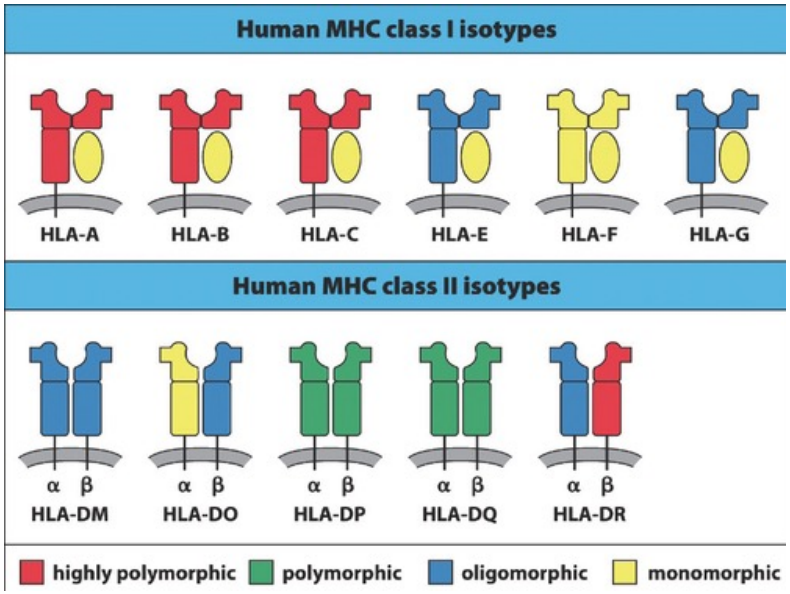
1. Basic principles of classical MHC I & II antigen processing & presentation

2. Antigen recognition of pathogens (applies also to tumours)

3. Study case: SARS CoV-2 antigen processing & MHC I presentation

Molecular Basis of MHC Recognition & MHC Polymorphism





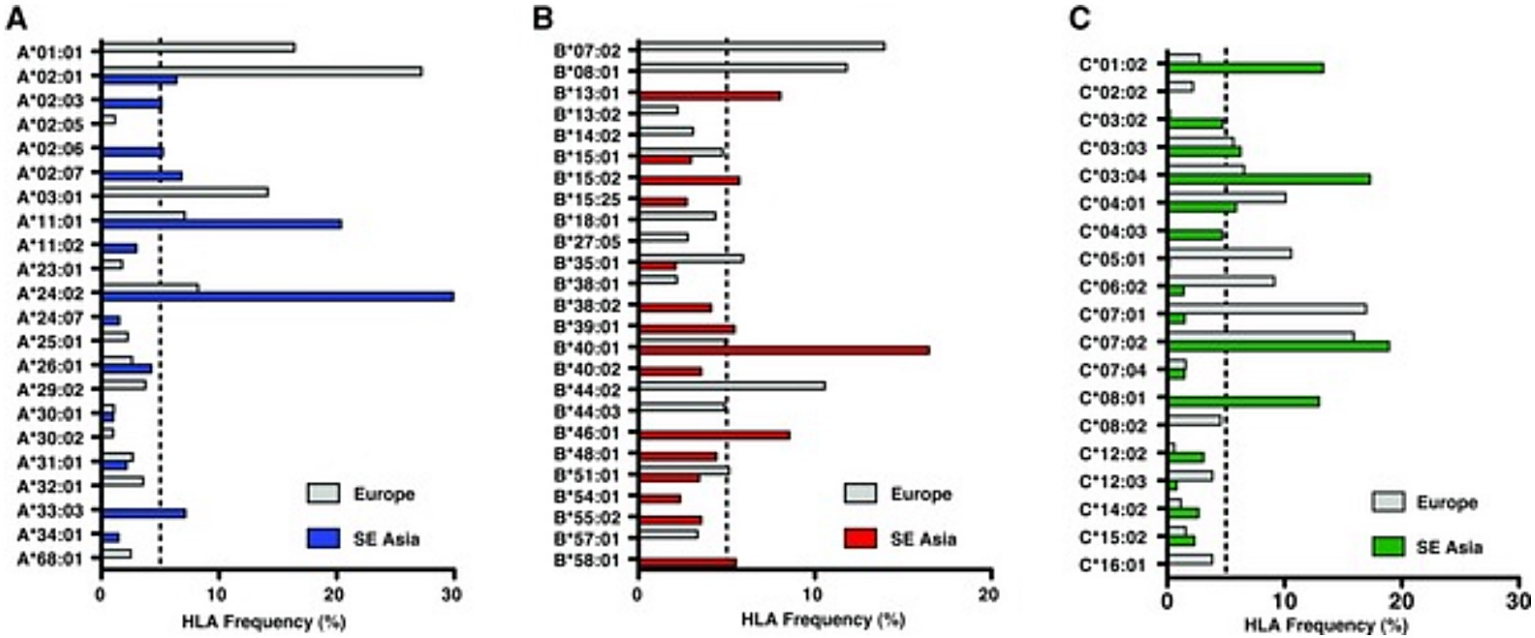
MHC:
Major histocompatibility complex class I & II

Antunes *et al.,* Current Topics in Medicinal Chemistry (2018) 18(26):2239-2255

How many different peptides can be presented by MHC I molecules ?

Do you know the major HLA (MHC I) alleles in the SE Asian population ?

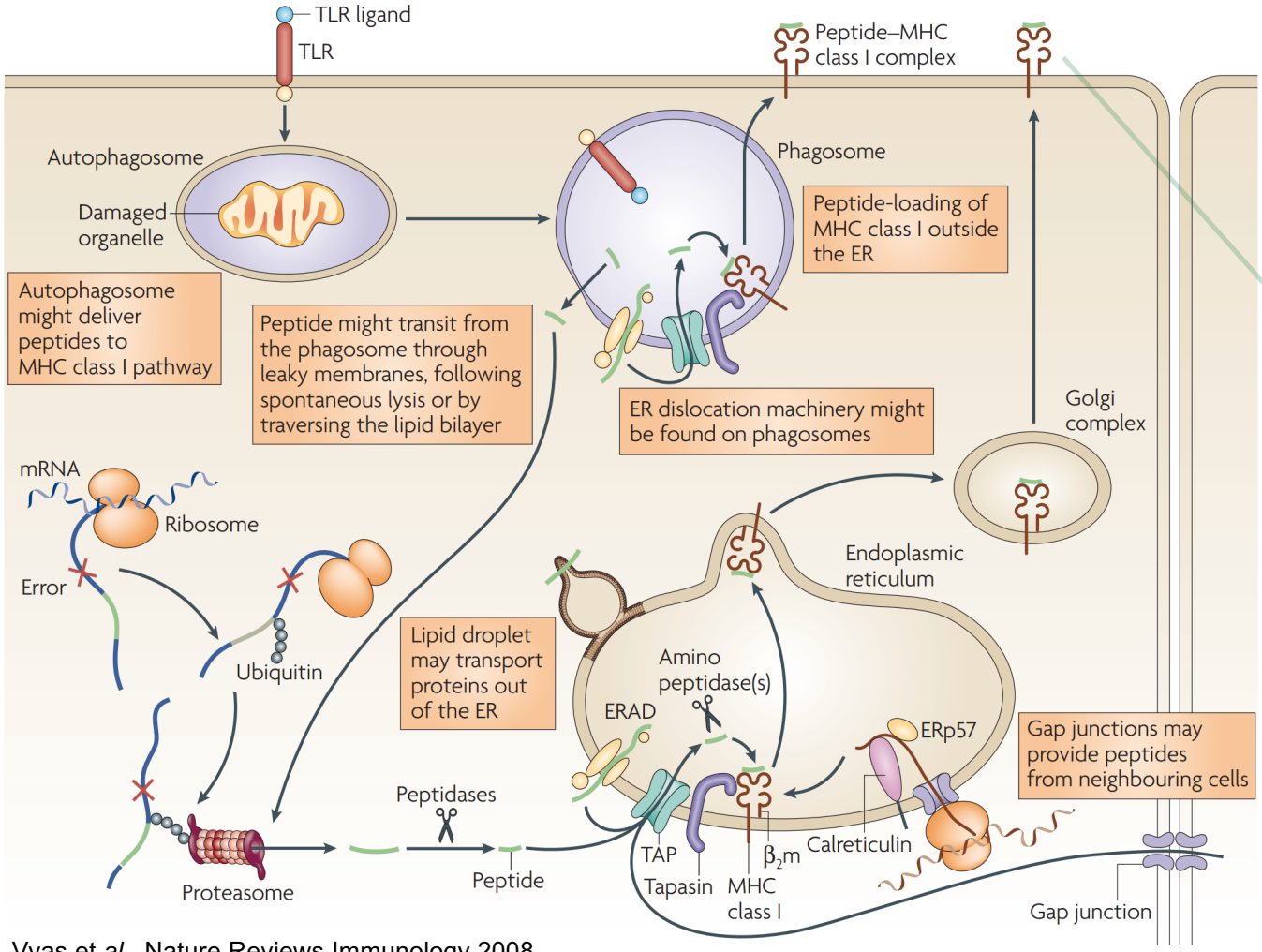
MHC I Alleles Across Different Populations



Frequency of HLA-A, HLA-B, and HLA-C alleles in the European and South East Asian population. (A) The allele frequencies for 22 HLA-A loci, based on data from NCBI (dbMHC anthropology resources alleles and haplotype frequencies website URL: http://www.ncbi.nlm.nih.gov/gv/mhc/) pooled from 95 published reports, were plotted for both European and South East Asian populations. Alleles with frequencies less than 2% were omitted. The dotted line indicates a 5% cut-off value. (B) The allele frequencies for 25 HLA-B loci are shown as in panel A. (C) The allele frequencies for 19 HLA-C loci are shown as in panel A.

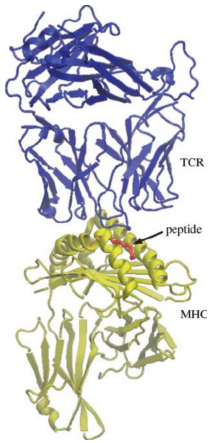
Chang *et al.,* European Journal of Immunology April 2013 43(4)

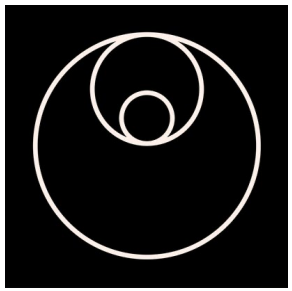
MHC-I Processing & Recognition of Antigens

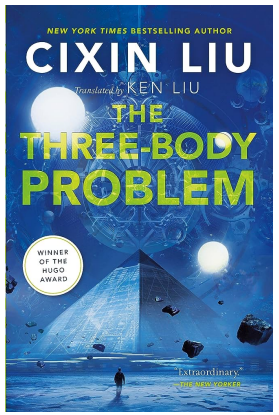


Vyas et *al.*, Nature Reviews Immunology 2008

CTL: Cytotoxic T lymphocytes (CD8 T-cells)
MHC: Major histocompatibility complex class I

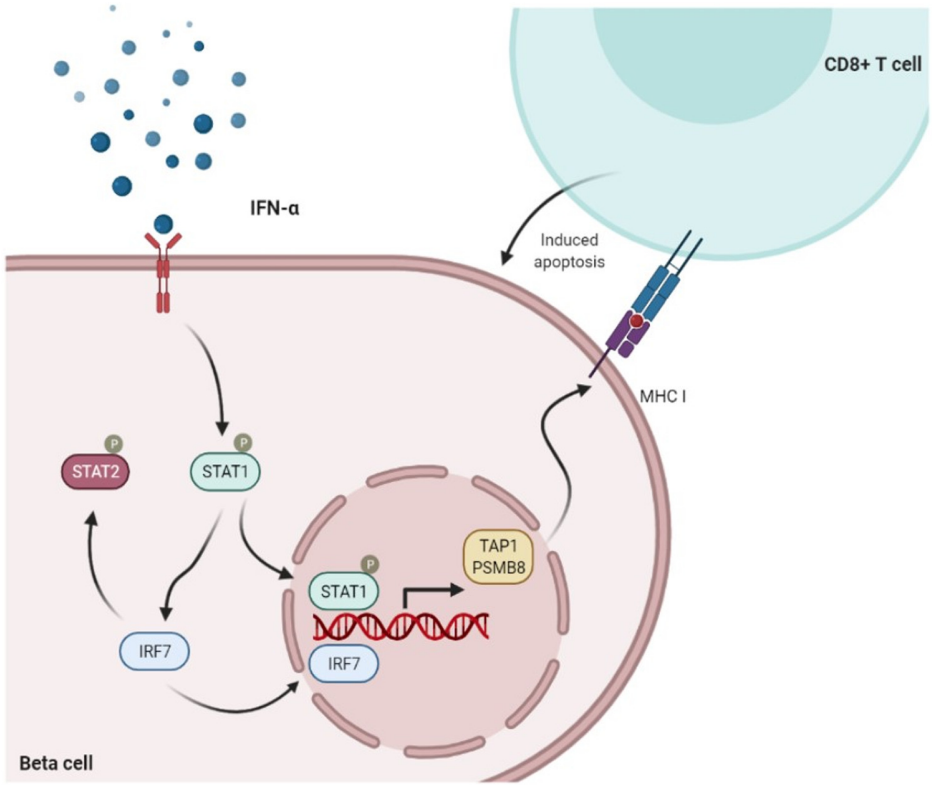








Interferons Stimulate Antigen Presentation



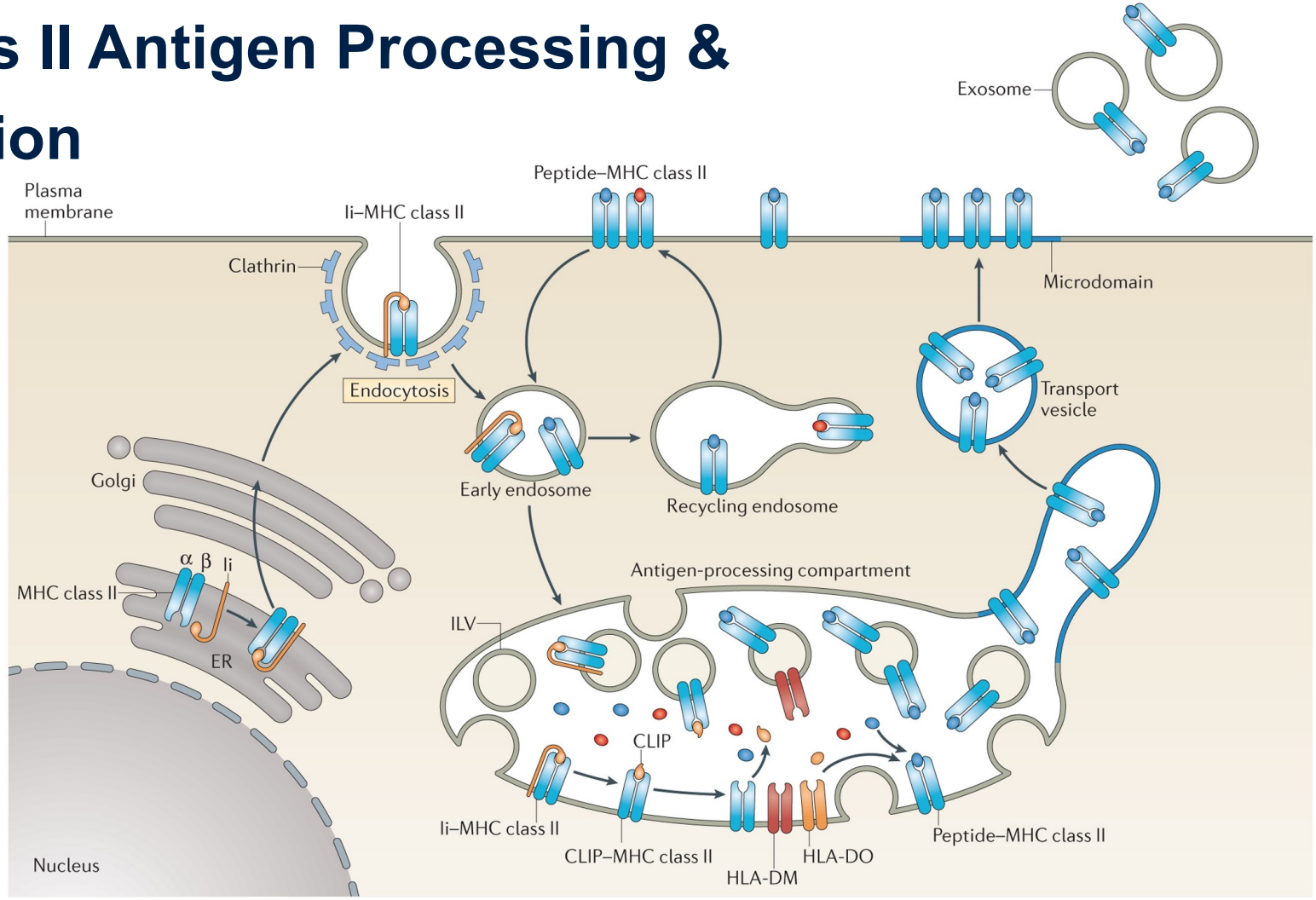
Interferon type I & II

TNFalpha

TLR signalling

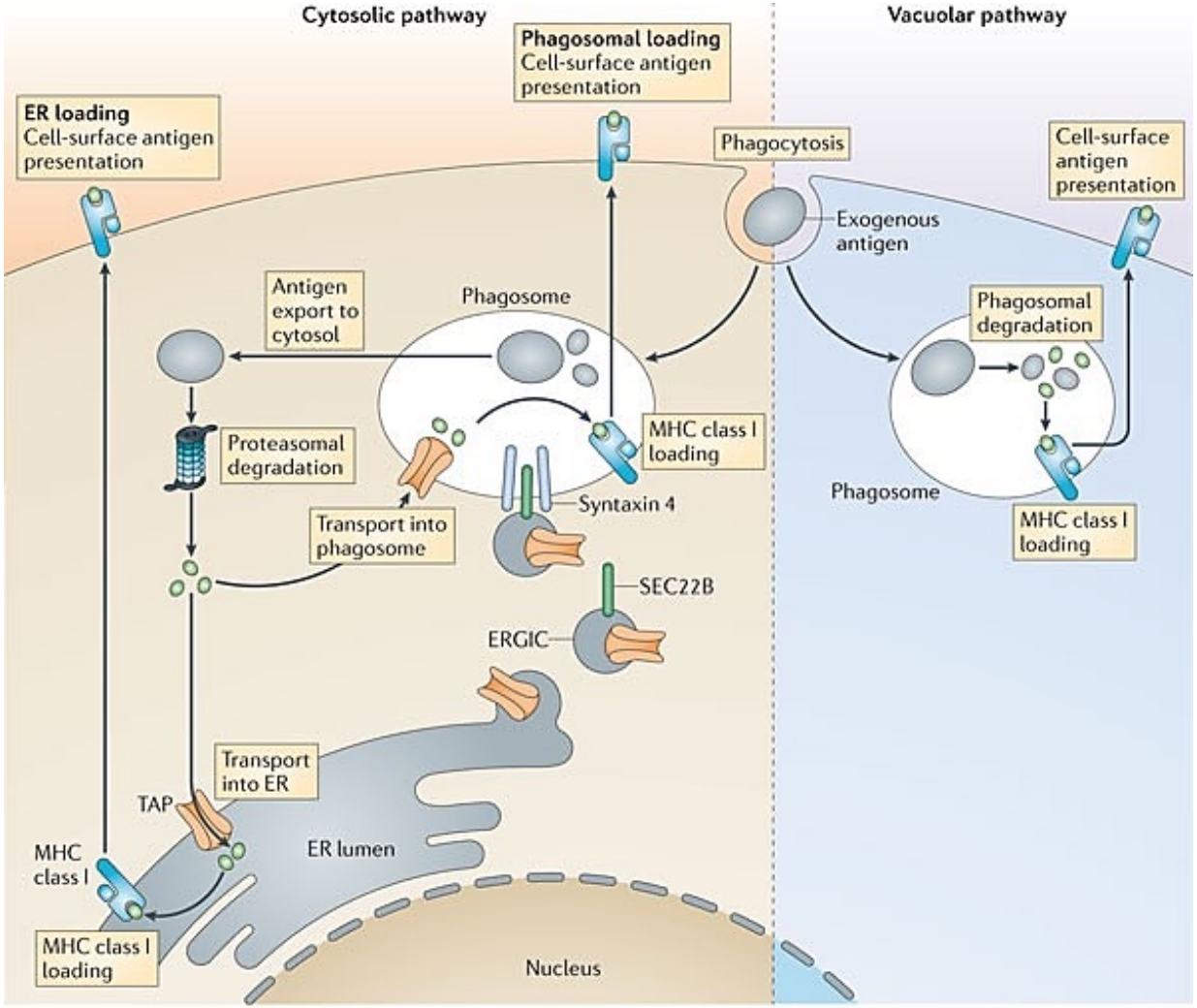
Jiang H, et al., Immunology. 2022; 166: 210–221. https://doi.org/10.1111/imm.13468

MHC class II Antigen Processing & Presentation



Roche et al., 2015 *https://doi.org/10.1038/nri3818* **Nature Reviews** | Immunology

Cross- Presentation



Nature Reviews | Immunology

Pathogens: Examples of viruses & bacteria – entero-pathogens

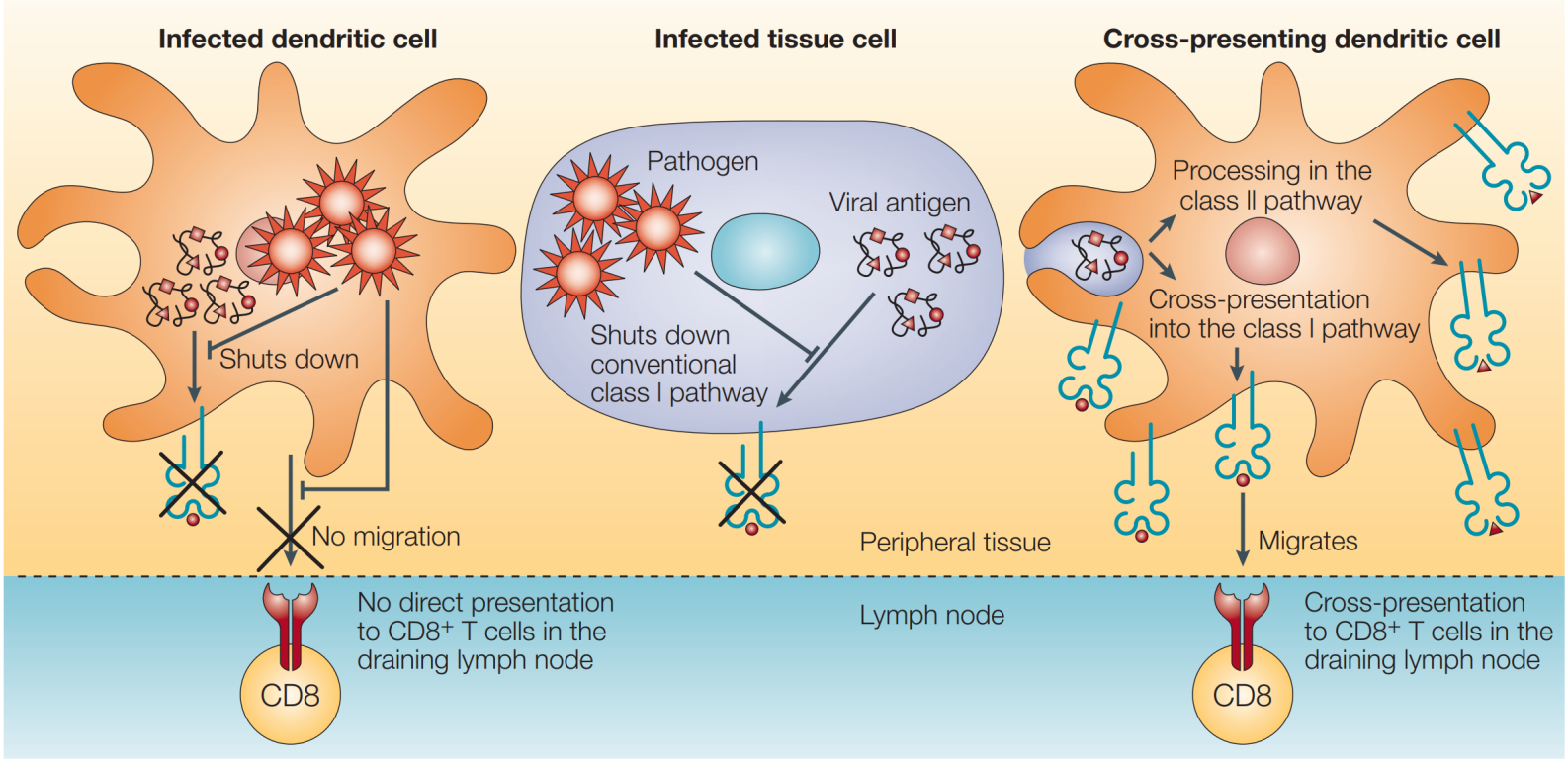
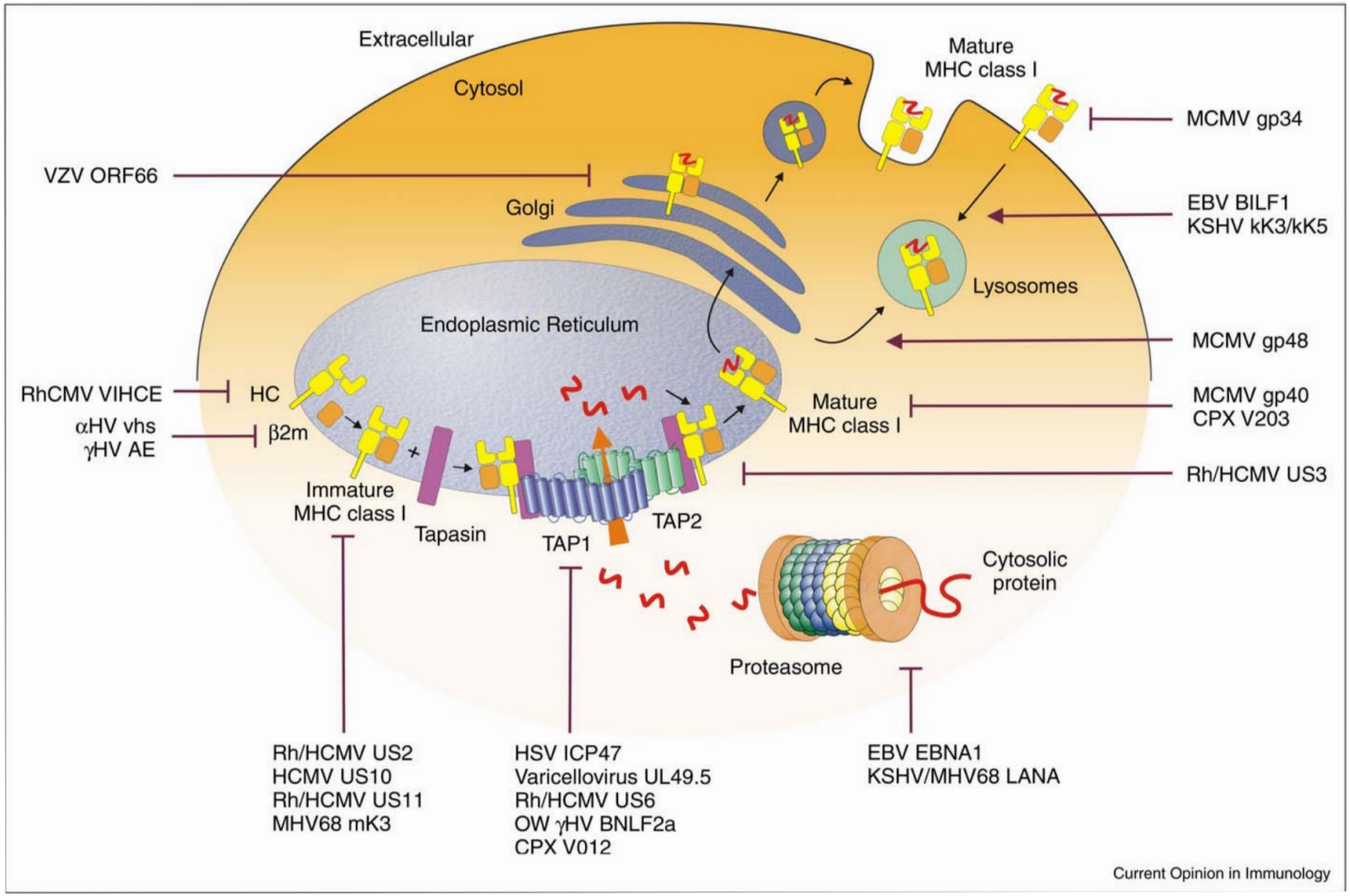


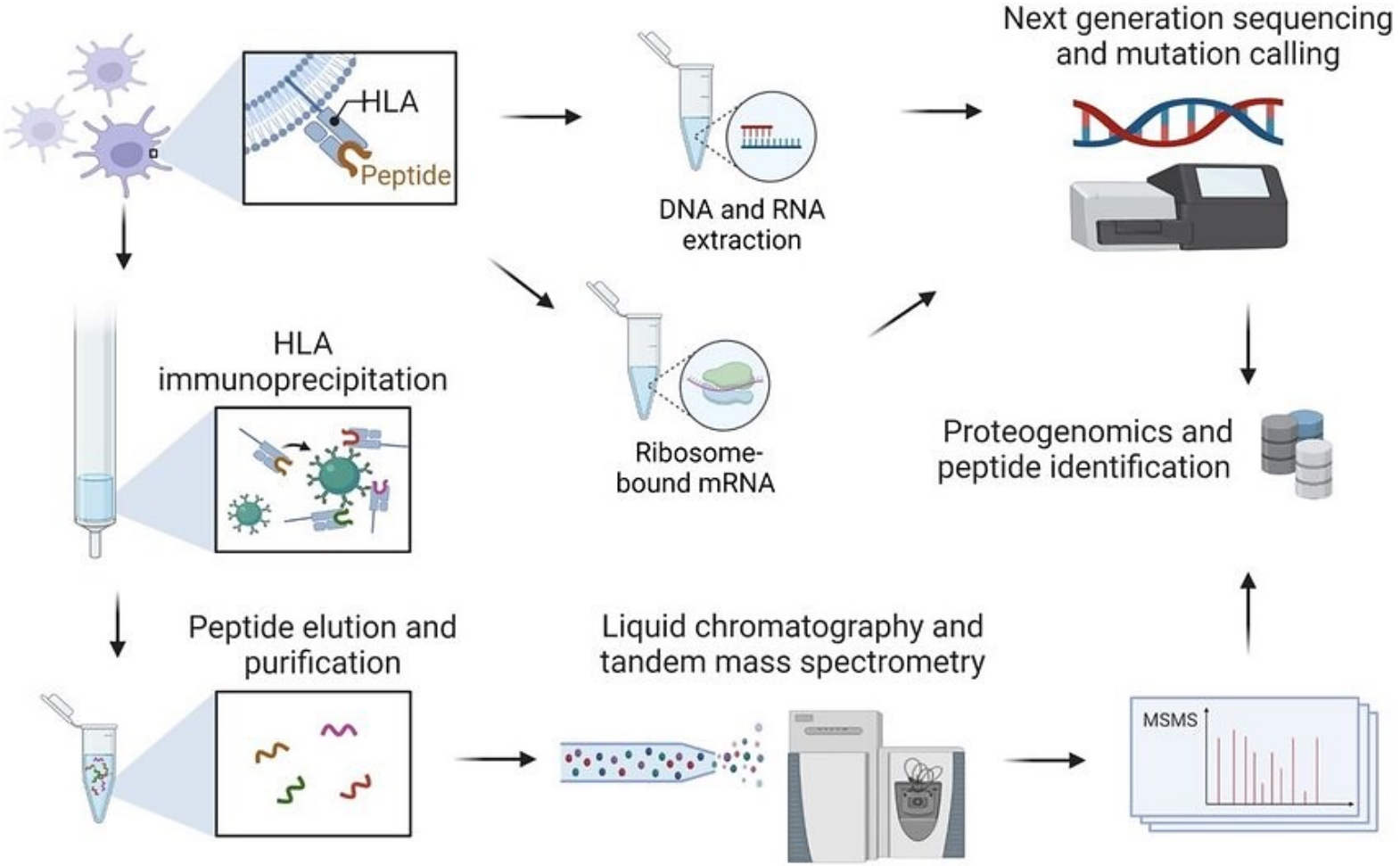
Figure 5 | **Viral subversion of dendritic cell function: cross-priming is required to generate CTL immunity.** Some viruses impair dendritic cell (DC) function during infection, for example by inhibition of MHC class I-restricted presentation or blocking migration of DCs. In these circumstances, viral inhibition of infected DCs prevents cytotoxic T lymphocyte (CTL) stimulation by the direct MHC class I pathway. However, cross-presentation of antigens that are derived from infected cells by uninfected DCs is likely to result in CTL immunity in the face of inhibitory mechanisms. Although virus-mediated inhibition of target-cell expression of MHC class I will also impair the effector phase of responses induced by cross-priming, this is unlikely to completely block recogniton by effector CTLs.

Which pathogens evade the immune system most efficiently?

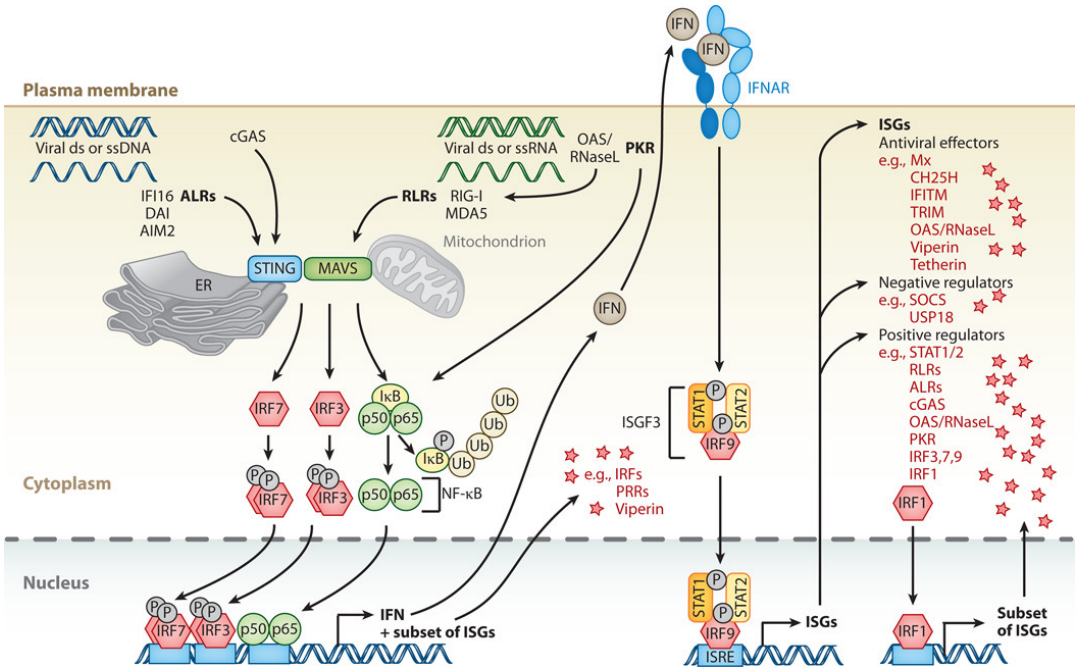
Herpes viruses and immune evasion

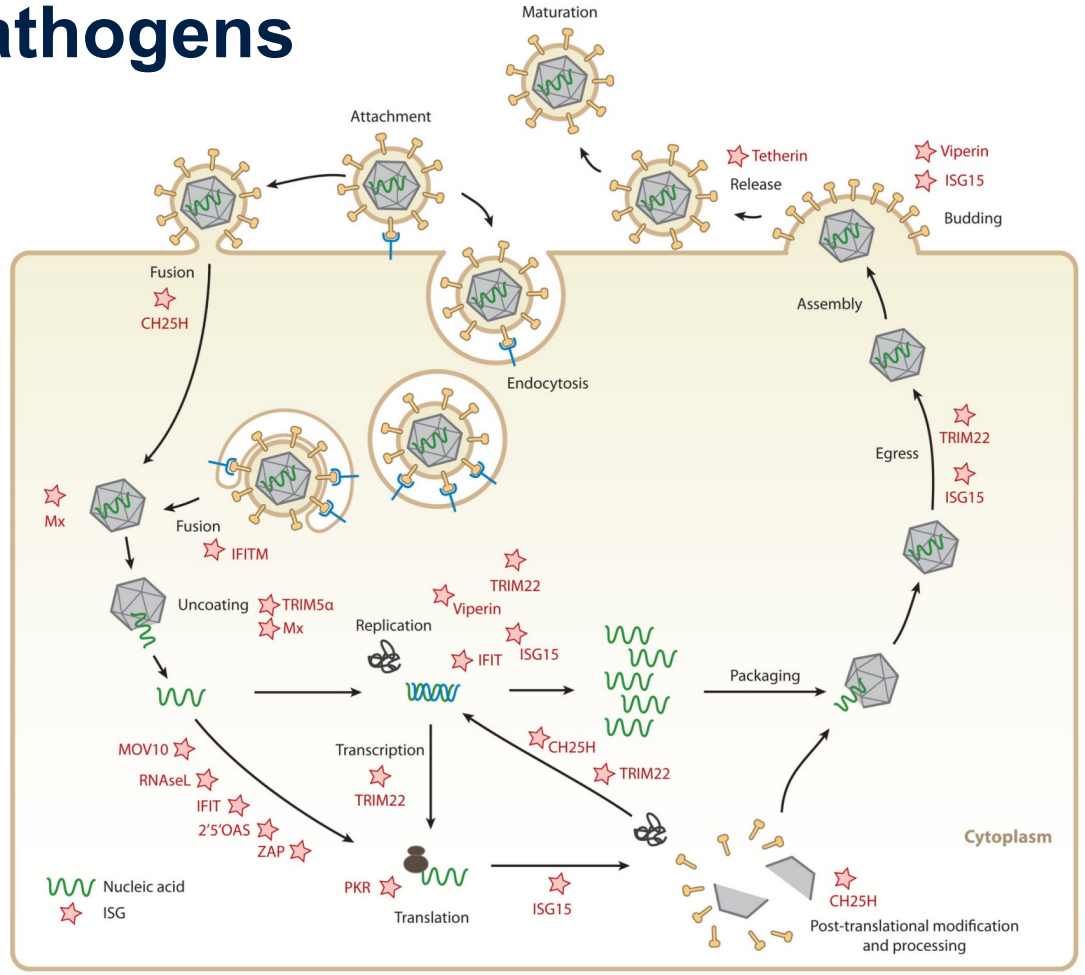


Immunopeptidomics to track pathogen antigen presentation



Interferon response against pathogens





Ag processing genes are ISGs
MHC I
TAP1/2

Schneider WM Annual Review of Immunology. 2014 ;32:513-545.

Mutations in SARS-CoV-2 CD8+ T cell nucleoprotein (NP) epitope flanking regions

Name	Epitope Sequence	HLA Restriction
NP_{9-17}	QRNAPRITF	B*27:05
$NP_{105-113}$	SPRWYFYYL	B*07:02

Peng et al., Nat Immunol 2020

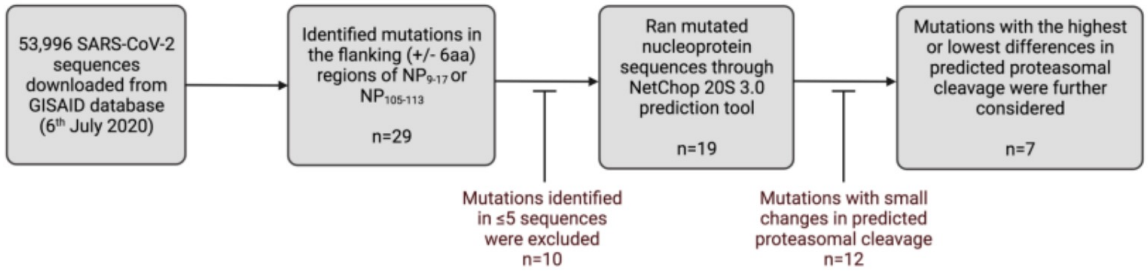
Dannielle Wellington



Suet Ling Felce



A



B

	Fold change in NetCHOP 20S prediction compared to WT														
Cleavage position	WT	D3L	G5E	P6L	Q7H	Q7K	G18V	G18S	G18C	G19V	G19R	P20L	P20S	D22G	D22Y
G5	1.00	1.05	3.11	0.88	3.31	0.26	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
P6	1.00	2.33	1.27	15.65	1.08	1.08	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
F17	1.00	1.45	1.00	1.00	1.00	1.00	1.00	1.06	1.04	0.94	0.88	1.06	1.03	1.00	1.00
G18	1.00	1.00	1.00	1.00	1.00	1.00	3.64	1.30	2.50	1.05	2.81	2.44	0.75	1.00	1.00
G19	1.00	1.00	1.00	1.00	1.00	1.00	1.52	1.27	1.92	4.36	6.68	0.78	1.62	1.28	1.97
P20	1.00	1.00	1.00	1.00	1.00	1.00	4.79	1.30	1.91	1.44	1.21	19.05	3.95	1.02	1.14
D22	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.01	1.29	1.09	0.40	1.21

C

	Fold change in NetCHOP 20S prediction compared to WT					
Cleavage position	WT	M101I	D103N	D103Y	A119S	A119V
M101	1.00	0.72	1.18	1.18	1.00	1.00
D103	1.00	0.93	0.21	1.48	1.00	1.00
L104	1.00	1.01	1.17	1.16	1.00	1.00
D103	1.00	1.00	1.00	1.00	1.00	1.00
A119	1.00	1.00	1.00	1.00	0.52	1.20

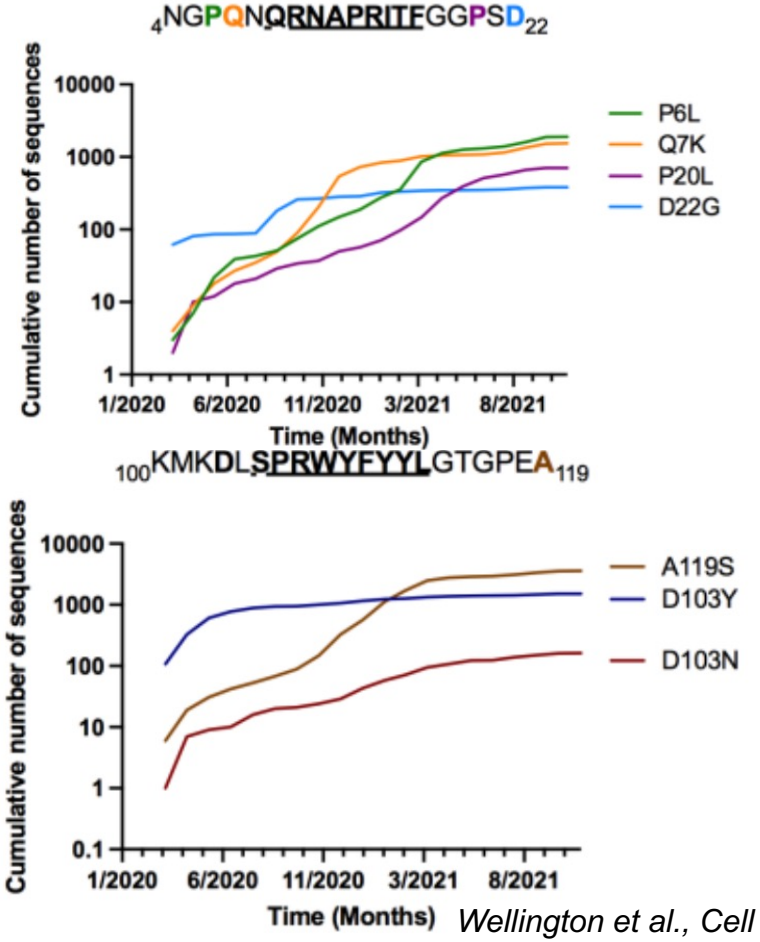
Increased cleavage
Decreased cleavage

Wellington et al., Cell Heliyon 2023

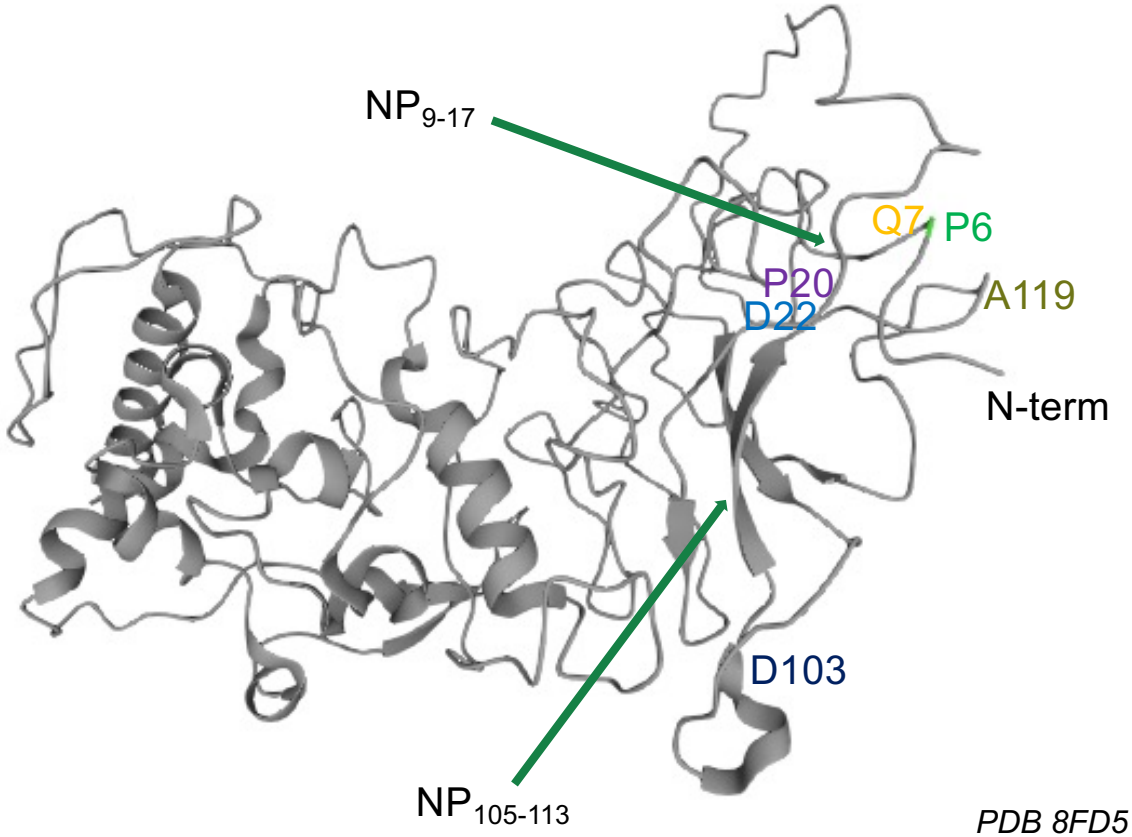
Mutations in SARS-CoV-2 CD8+ T cell nucleoprotein (NP) epitope flanking regions



Dannielle Wellington



Wellington et al., Cell Heliyon 2023



PDB 8FD5

Casasanta Microscopy and Microanalysis 2023

Mutations in flanking regions affect proteasomal digestion and epitope generation of SARS-CoV-2 CD8+ T cell epitopes

Proteasome Processing Assay

Peptide epitope precursors

Proteasome digestion in vitro
20S / i20S

LC-MS/MS
Label-free quantitation (LFQ)



Evosep LC
timsTOF



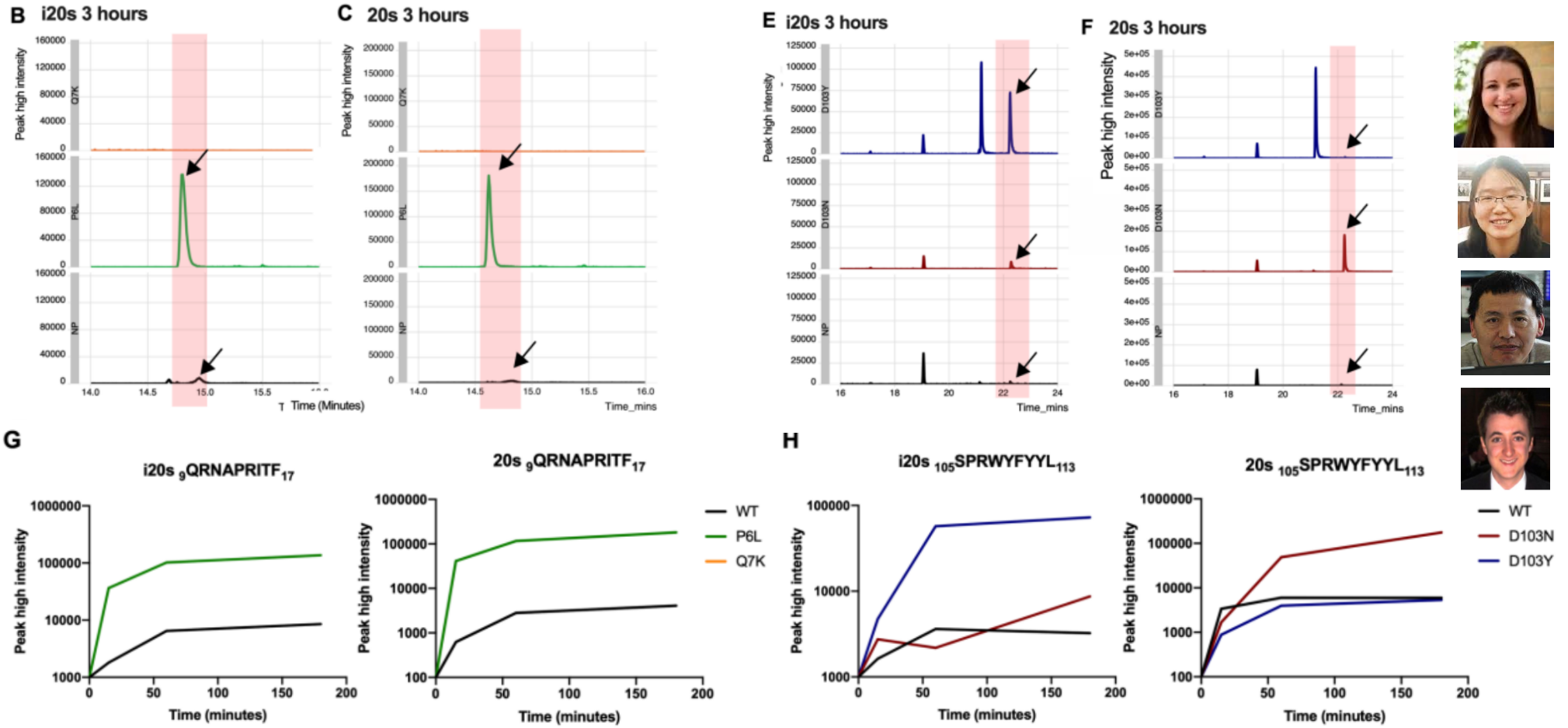
Wellington et al., Cell Heliyon 2023

Dannielle Wellington
Zixi Yin (Tracey)
Zhanru Yu
Simon Davis
Zhu Liang



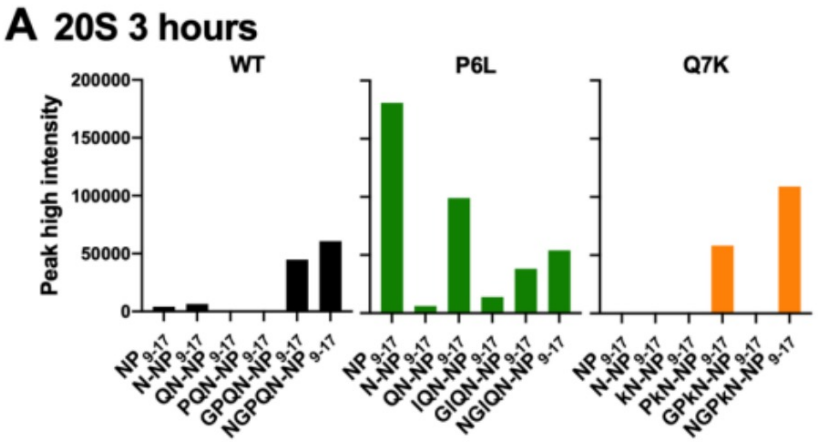
Mutations in flanking regions affect proteasomal digestion and epitope generation of SARS-CoV-2 CD8+ T cell epitopes

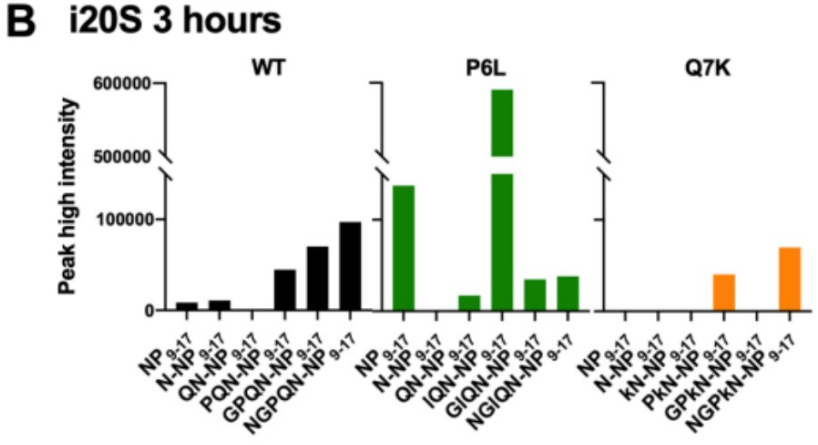
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Zhanru Yu
Simon Davis



Wellington et al., Cell Heliyon 2023

Flanking mutations alter N-terminally extended proteasomal products





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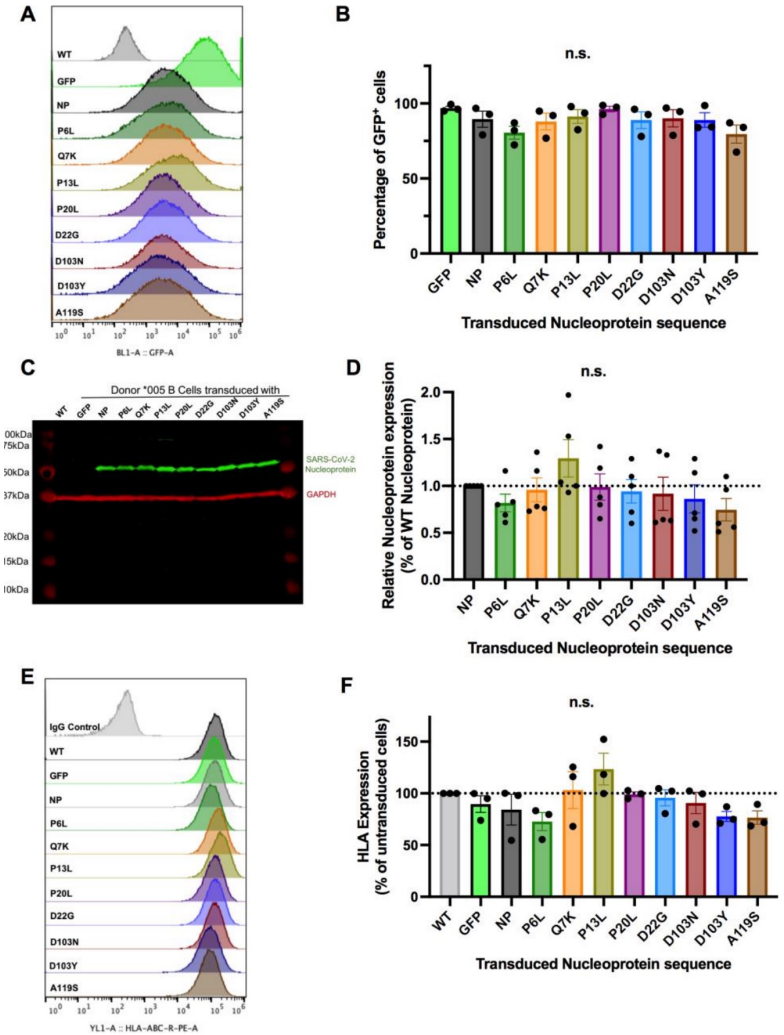




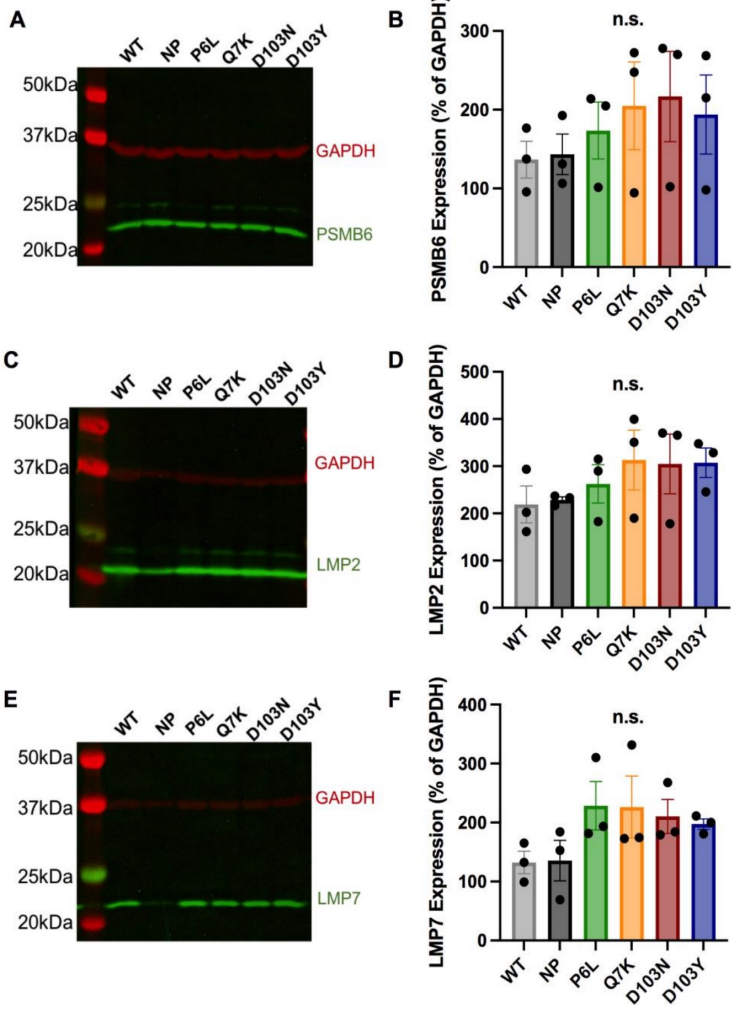




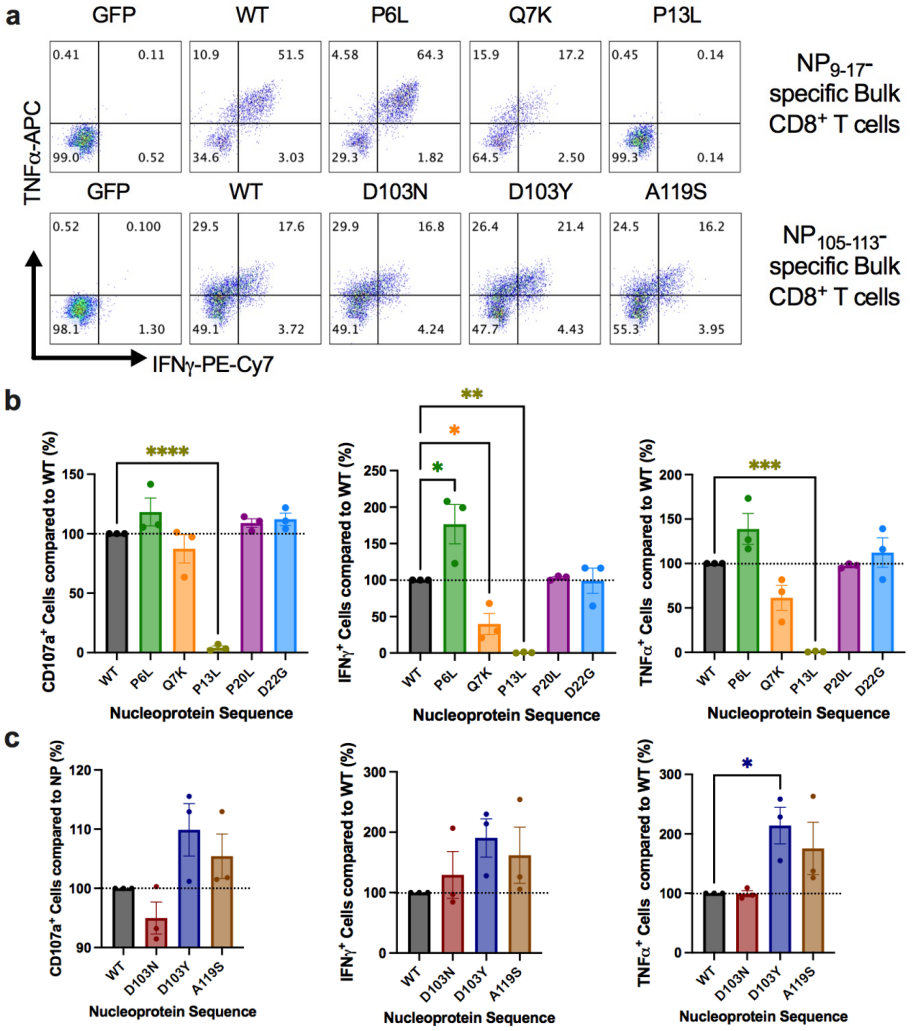
Characterisation of nucleoprotein-transduced B Cells

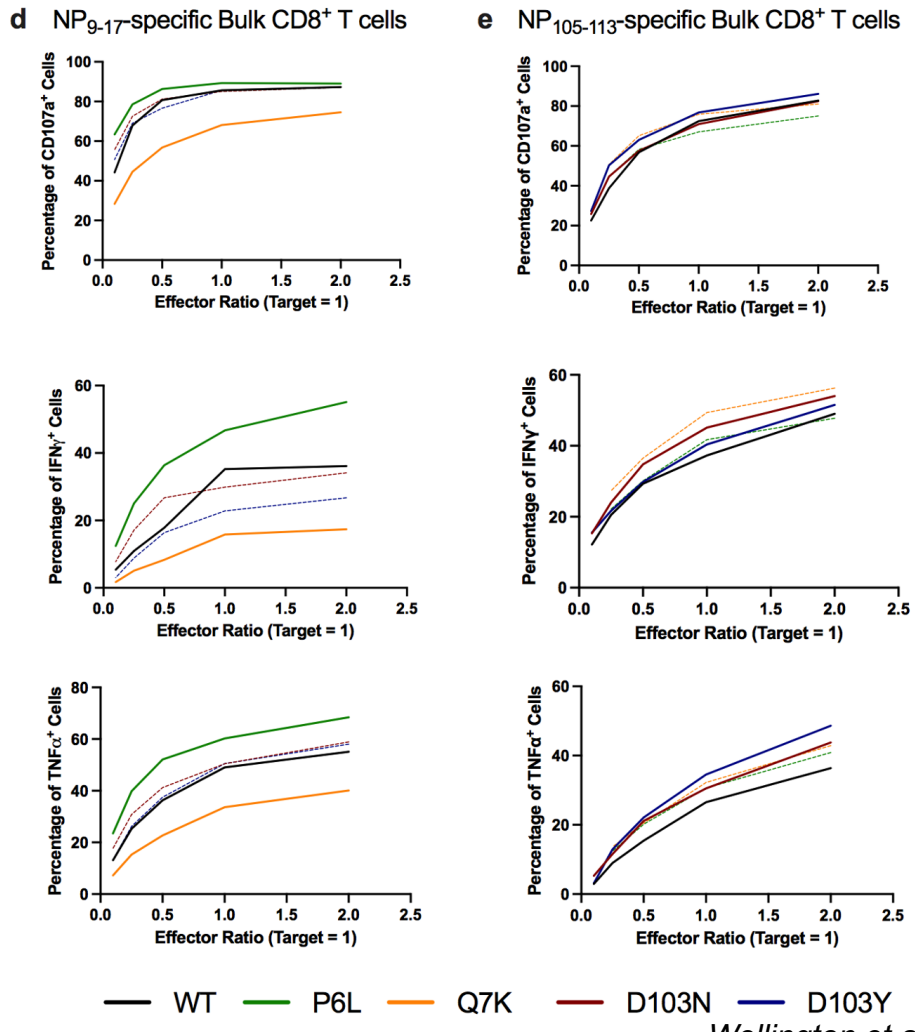


Proteasomal subunit expression in B cell lines

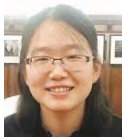


Mutations in flanking regions of epitopes alter CD8+ T Cell responses





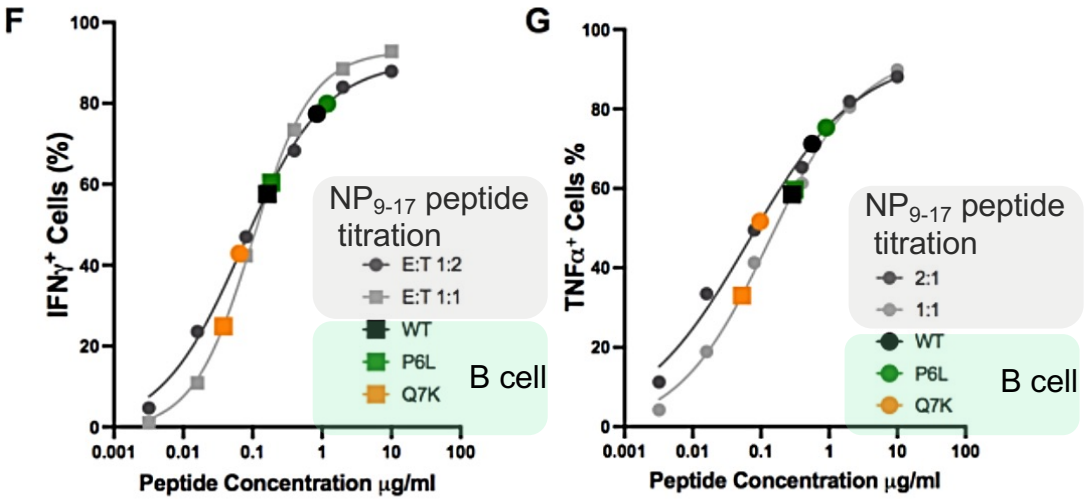
Dannielle Wellington
Tracey
Zhanru Yu

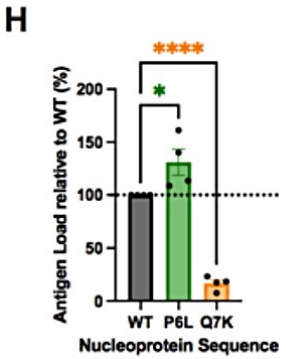





Wellington et al., Cell Heliyon 2023

Flanking mutations affect cell surface antigen load and CD8+ T-cell activation





Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

SARS-CoV-2 mutations affect antigen processing by the proteasome to alter CD8+ T cell responses

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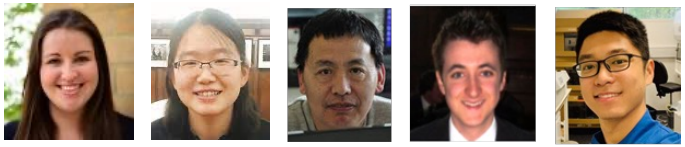
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Wellington et al., bioRxiv 2022
Wellington et al., Cell Heliyon 2023

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Tracey
Zhanru Yu
Simon Davis





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